



Practical Exam

1-Audio Compression Project: MP3 Encoder Implementation

Objective: Implement a basic MP3-like audio compression system

Steps:

1. **Audio Input Handling**
 - Create a raw PCM audio file (e.g., WAV format). That contains noise and silence
 - Plot audio signal without noise
 - Plot audio signal that includes noise and silence
 2. **Transform to Frequency Domain**
 - Apply Short-Time Fourier Transform (STFT) or MDCT.
 - Split the audio into frequency bands.
 3. **Quantization & Bit Allocation**
 4. **Encoding**
 - Use Huffman or run-length encoding to compress quantized values.
 5. **Testing & Evaluation**
 - Compare original vs. decompressed audio.
 - Use Signal-to-Noise Ratio (SNR)
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2-Video Compression Project:

Objective: Implement a basic video compression.

Steps:

1. **Video Input Handling**
 - Create video frame-by-frame
 - Convert frames to YUV color space.
2. **Frame Type Decision**
 - Choose I-frames and P-frames (e.g., every 10th frame is an I-frame).
3. **Intra-frame Compression (I-frame)**
 - Apply DCT on 8x8 blocks.
 - Quantize DCT coefficients.
 - Apply zig-zag scan and run-length encoding.
4. **Inter-frame Compression (P-frame)**
 - Perform motion estimation (e.g., block matching).
 - Compute motion vectors.
 - Encode motion vectors and residuals.
5. **Entropy Coding**
 - Use Huffman or Arithmetic Coding on motion vectors and residuals.
6. **Bitstream Formation**
 - Package frames into a bitstream with headers and frame type indicators.
7. **Testing & Evaluation**
 - Compare original and decoded video.
 - Measure compression ratio and PSNR (Peak Signal-to-Noise Ratio).